

Automated CV Screening System

Brief Setup Documentation

1. Overview

This system automates CV intake, analysis, role matching, and notification using **n8n** as the orchestration layer.

It accepts CV uploads via a form, extracts and analyzes CV content using AI, compares candidates against open roles, and notifies relevant parties based on match results.

The workflow is designed to be **deterministic, auditable, and easily extendable, with a** clear separation between validation, AI reasoning, and business rules.

2. Architecture & Flow Summary

1. Candidate submits CV via a web form
2. Server-side validation ensures file integrity and safety
3. CV is stored in Supabase Storage
4. CV content is extracted using Mistral OCR
5. AI agent evaluates CV against open roles from Airtable
6. Matching logic is applied
7. Results trigger either:
 - HR notification (Slack), or
 - Polite rejection email to the candidate

3. Tools & Services Used

Component	Tool / Service	Purpose
Workflow orchestration	n8n	Primary automation platform
Form handling	n8n Form Trigger	CV intake
File storage	Supabase Storage	Persist uploaded CVs
CV parsing / OCR	Mistral AI	Extract text from CVs
AI reasoning	Mistral LLM (LangChain Agent)	CV analysis and role matching

Roles database	Airtable	Store job requirements
HR notification	Slack	Internal candidate alerts
Candidate communication	Gmail	Application updates

4. Validation & Data Handling

- **Frontend validation** (form): improves user experience and prevents obvious errors
- **Server-side validation** (code node):
 - Enforces MIME types (PDF/DOCX)
 - Caps file size (5MB)
 - Guards against missing or malformed uploads
- Validation is intentionally duplicated to protect against bypassed client-side controls.

Uploaded files are renamed and normalized before storage to ensure predictable paths and safe filenames.

5. AI Matching Logic

- CV metadata (name, email) is treated strictly as a non-professional context.
- Skills, experience, and qualifications are extracted **only from CV text**.
- Open roles are retrieved exclusively via an Airtable tool to prevent hallucination.
- For each role:
 - Skill match percentage is calculated
 - Experience shortfalls reduce the score proportionally
- Only the single highest-scoring role is considered.
- AI output is enforced as strict JSON.

Match statuses are intentionally limited to:

- `matched`
- `no_match`

This aligns directly with the test requirements and keeps downstream logic simple.

6. Business Rules

- A minimum match score threshold ($\geq 75\%$) is enforced at the workflow level.
 - AI provides analysis; the final qualification is determined deterministically.
 - Human review is required before any hiring decision.
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7. Notifications

- **Matched candidates:**
HR is notified via Slack with candidate details and match score.
- **Unmatched candidates:**
A polite, automated email is sent to the applicant.

This ensures timely communication while preserving a positive candidate experience.

8. Storage & Auditability

- CV files are stored in Supabase Storage.
 - Matched candidate metadata is saved in a Supabase table.
 - Stored references allow later auditing, reprocessing, or review.
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9. Additional Files & Configurations Used

External Configurations

- **Supabase**
 - Storage bucket for CV uploads
 - Table for matched candidates
- **Airtable**
 - Roles table with structured job requirements
- **Slack**
 - Channel for HR notifications
- **Gmail**
 - OAuth credentials for candidate emails
- **Mistral AI**
 - OCR and language model credentials

n8n Configuration

- Custom code node for server-side validation
- Structured Output Parser for AI response enforcement

- Tool-enabled AI agent (LangChain)
 - Environment credentials stored securely in n8n
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10. Known Limitations & Future Improvements (Optional)

- Add role weighting (e.g., critical vs optional skills)
 - Introduce async processing for large files
 - Add candidate scoring history and analytics
 - Implement retry logic for transient AI failures
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Final Note

The workflow prioritizes **clarity, safety, and explainability** over over-engineering. AI is used where judgment is required, while deterministic rules control outcomes.